Project Instructions

AUSTRIA VOTER TRANSITIONS | 2019–2024 EUROPEAN PARLIAMENT ELECTIONS

Thank you for taking the time to review this project. Please read these brief instructions first; they explain the file structure and how to reproduce the analysis.

1. Project Overview

This R project uses Austria as a case study to estimate potential voter transitions between European Parliament political groups from the 2019 to the 2024 European Parliament elections.

The analysis is based on electoral data from Austrian national political parties that contested the 2019 and 2024 European Parliament elections. Votes for each national party are classified according to the European Parliament political group to which the party belonged in the respective election year. The resulting political-group-level data are used to estimate potential voter movements between European Parliament political groups and abstention, while allowing the estimates to be interpreted in relation to the corresponding Austrian national parties.

Rather than focusing solely on domestic dynamics, the analysis and charts are structured around European Parliament political groups. Each Austrian party is mapped to its European affiliation, maintaining a clear connection between European-level transitions and national parties.

Voter transitions are estimated using the Voter Transition Rates (VTR) methodology and accompanying R code developed by Ioannis Andreadis and Theodore Chadjipadelis. The code is provided by the Laboratory of Applied Political Research, Department of Political Sciences, Aristotle University of Thessaloniki, Greece. The methodology is described in Andreadis and Chadjipadelis (2009), “A Method for the Estimation of Voter Transition Rates,” Journal of Elections, Public Opinion and Parties.

2. Project File Structure

All files required to review and reproduce the analysis are included in the project folder. Please keep them together when running the project.

```
Austria_Voter_Transitions/
├──── Austria_Electoral_Analysis_README.pdf
├──── Austria_Voter_Transitions.Rproj
├──── analysis.R
├──── VTR.R
├──── austria_2019.csv
├──── austria_2024.csv
├──── austria_map.geojson
├──── Austria_Electoral_Analysis.Rmd
├──── Austria_Electoral_Analysis_RMarkdown.pdf
└──── Austria_Electoral_Analysis.pdf
```

File	Purpose
Austria_Electoral_Analysis_README.pdf	This document: project overview and step-by-step instructions.
Austria_Voter_Transitions.Rproj	RStudio project file. Open first to set the working directory automatically.
analysis.R	Main analysis: data preparation, checks, VTR estimation, exports, and visualizations.
austria_2019.csv	Input data for the 2019 European Parliament election.
austria_2024.csv	Input data for the 2024 European Parliament election.
VTR.R	Supporting R functions for voter-transition estimation, including multirate().
Austria_Electoral_Analysis.Rmd	R Markdown file containing the electoral analysis, including voter transition estimates, visualizations, and regional electoral changes in Austria between the 2019 and 2024 European Parliament elections.
Austria_Electoral_Analysis_RMarkdown.pdf	Rendered R Markdown document containing the R code, voter-transition estimates, tables, visualizations, and regional electoral maps.
Austria_Electoral_Analysis.pdf	Research report presenting the study's objectives, theoretical framework, methodology, findings, and conclusions regarding electoral changes in Austria between the 2019 and 2024 European Parliament elections.

austria_map.geojson	Geographical data containing the boundaries of Austrian NUTS 2 regions, used to create maps illustrating regional changes in FPÖ and ÖVP electoral support.

3. How to Open and Run the Project

Step 1 — Open RStudio

In RStudio, choose File > Open Project... and select Austria_Voter_Transitions.Rproj, or double-click the.Rproj file.

Step 2 — Open the main script

Once the project is open, choose File > Open File... and open analysis.R, or open it from the Files pane. No manual setwd() command is required. Opening Austria_Voter_Transitions.Rproj sets the project directory automatically.

Step 3 — Check required packages

At the beginning of analysis.R, immediately above the library () commands, there is a commented install.packages() command. If the packages are already installed, leave this line commented and continue to Step 4.

Step 4 — Install packages if needed

If any required package is missing, remove the # at the beginning of the install.packages() line. Place the cursor on that line and press Ctrl+Enter or click Run. This executes the installation command. After installation, the # can be added again because package installation is normally required only once on a computer.

Step 5 — Run the analysis

Run the analysis.R from top to bottom. To execute one line, place the cursor on it and press Ctrl+Enter or click Run. To execute several commands, select the relevant lines and press Ctrl+Enter or click Run. To execute the complete script at once, click Source in RStudio.

4. Required R Packages

The project uses tidyverse, scales, and ggalluvial. The relevant lines are located at the beginning of analysis.R:

```
# Install only if necessary:
# install.packages(c("tidyverse", "scales", "ggalluvial", "sf", "rmarkdown"))

# Load packages for this session:
library(tidyverse)
library(scales)
library(ggalluvial)
```

The library () lines should remain active because they load the packages into the current R session.

5. Working Directory and Input Files

Opening Austria_Voter_Transitions.Rproj in RStudio automatically sets the project directory.

The analysis uses relative file paths to access the required input files:

```
austria2019 <- read_csv("austria_2019.csv")

austria2024 <- read_csv("austria_2024.csv")

source("VTR.R")

austria_map <- st_read("austria_map.geojson", quiet = TRUE)
```

This allows the analysis to be reproduced on different computers, provided that all required files are kept in the project directory.

6. Analysis Workflow

1. Load the required R packages.
2. Import the 2019 and 2024 Austrian election datasets.
3. Inspect variable names and prepare electorate-size and transition matrices.
4. Convert election outcomes into proportions of registered voters.
5. Perform consistency, range, and row-sum checks.
6. Source the voter-transition functions from VTR.R.
7. Estimate voter transitions using multirate().

8. Export the aggregate transition matrix.
9. Reshape the transition matrix into long format for interpretation.
10. Identify selected major transitions and retention patterns.
11. Produce visualizations of estimated voter retention and selected voter movements.
12. Import and process Austrian NUTS 2 geographical data using the sf package.
13. Calculate regional changes in ÖVP electoral support between 2019 and 2024.
14. Create a regional map of changes in ÖVP electoral support and export the corresponding results.
15. Calculate regional changes in FPÖ electoral support between 2019 and 2024.
16. Create a regional map of changes in FPÖ electoral support and export the corresponding results.

7. VTR.R

VTR.R does not need to be opened or executed separately. The main script loads it automatically using source("VTR.R"). It contains supporting functions for voter-transition estimation, including multirate().

8. Methodological Note

The voter-transition results are ecological estimates derived from aggregate regional election data. They are not direct observations of individual voters and should be interpreted as estimated aggregate transition patterns rather than individual-level voting histories.

9. Political-Group Interpretation

FPÖ is associated with Identity and Democracy (ID) in the 2019 data and Patriots for Europe (PfE) in the 2024 data. Accordingly, an estimated ID → PfE transition represents estimated continuity among FPÖ voters across a change in European political-group affiliation; it should not be interpreted as retention within an unchanged European group.

10. Quick Start

1. Read Austria_Electoral_Analysis_README.pdf.
2. Open Austria_Voter_Transitions.Rproj in RStudio.

3. Open analysis.R.
4. Install the required R packages if necessary.
5. Run analysis.R from top to bottom.

11. How to Explore the Project

- For a comprehensive overview of the research objectives, theoretical framework, methodology, findings, and conclusions, please refer to Austria_Electoral_Analysis.pdf.
- For the computational analysis, including R code, voter-transition estimates, tables, visualizations, and regional electoral maps, please refer to Austria_Electoral_Analysis_RMarkdown.pdf.
- The corresponding R Markdown source file, Austria_Electoral_Analysis.Rmd, is included for reproducibility.
- The main analysis script, analysis.R, contains the complete R code, while VTR.R provides the supporting voter-transition estimation functions.
- The accompanying CSV datasets and GeoJSON file provide the necessary inputs for reproducing the analysis.